

Problem-Solution Fit

Date	6 July 2026
Team ID	XXXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] a. University and high school students b. Self-taught programmers and online learners	6. CUSTOMER LIMITATIONS [CL] a. Budget: Low (students cannot easily afford private tutoring) b. Devices: Mostly limited to basic laptops or smartphones c. Other: Short attention spans and low patience due to academic stress	5. AVAILABLE SOLUTIONS PROS & CONS [AS] a. Human Tutors: <i>Pros:</i> Personalized. <i>Cons:</i> Expensive, not available 24/7. b. Generic AI (ChatGPT): <i>Pros:</i> Fast. <i>Cons:</i> Lacks emotional empathy, gives robotic answers that can overwhelm beginners. c. Online Forums (StackOverflow): <i>Pros:</i> Detailed. <i>Cons:</i> Can be toxic to beginners, slow response times.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]	9. PROBLEM ROOT / CAUSE [RC]	7. BEHAVIOR + ITS INTENSITY [BE]

- a. **Pains:** Getting stuck on complex topics, feeling overwhelmed by errors, and lacking personalized mentorship.
- b. Very high (occurs daily or weekly during active study sessions)

Traditional learning systems and standard AI tools lack **emotional intelligence**. They provide raw facts but fail to address the student's mental block or emotional distress, which is the real barrier to learning

- a. **Behavior:** Procrastination, "rage-quitting" study sessions, developing imposter syndrome.
- b. **Intensity:** High (can lead to dropping out of courses or severe academic anxiety).

3. TRIGGERS TO ACT [TR]

- a. a. Encountering a persistent bug/error that won't go away
- b. Failing an exam or struggling with an assignment deadline
- c. Feeling intense frustration leading to the urge to drop a course

10. YOUR SOLUTION [SL]

Emotion Detection & Learning Support Engine: An AI-powered web app that uses machine learning (BiLSTM & BERT) to detect a student's exact emotional state from their text. It then uses the Gemini API to deliver a personalized, empathetic learning strategy that addresses both their academic problem *and* their emotional wellbeing.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE: Searching Google frantically, posting on Reddit/Discord, watching YouTube tutorials.

4. EMOTIONS BEFORE / AFTER [EM]

- a. **Before:**
Frustrated,
Anxious,
Confused,
Overwhelmed,
Lonely.
- b. **After:** Supported,
Relieved,

OFFLINE: Searching Google frantically, posting on Reddit/Discord, watching YouTube tutorials.

Confident,
Understood.